

Course Code: ML402

Course Name: Machine Learning

Semester: Fall 2026

Course Objectives:

Learn the principles, algorithms, and applications of Machine Learning for solving real-world problems.

Course Content:

UNIT 1: Introduction to Machine Learning

- What is Machine Learning
- Types of Learning
- Data Preprocessing
- Feature Engineering
- Model Evaluation

UNIT 2: Supervised Learning

- Linear Regression
- Logistic Regression
- K-Nearest Neighbors
- Decision Trees
- Random Forest

UNIT 3: Unsupervised Learning

- Clustering
- K-Means Algorithm
- Hierarchical Clustering
- Dimensionality Reduction
- Principal Component Analysis

UNIT 4: Neural Networks and Deep Learning

- Perceptron
- Artificial Neural Networks
- Activation Functions
- Backpropagation
- Introduction to Deep Learning

UNIT 5: Advanced Machine Learning

- Support Vector Machines
- Ensemble Learning
- Reinforcement Learning
- Model Deployment
- Ethics in AI and ML

Suggested Books:

- Tom Mitchell – Machine Learning
- Christopher Bishop – Pattern Recognition and Machine Learning
- Aurélien Géron – Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

Course Outcomes:

- Understand machine learning concepts and workflows.
- Build supervised and unsupervised models.
- Understand neural networks and deep learning.
- Apply machine learning techniques to practical problems.